

Bounded Logical Homing

Release, seek, and establish one volatile origin

E0 COPIED REPLAY — NO SENSOR, COIL, MOTOR, OR POSITION CLAIM

Lesson 050 — component
Prerequisite Lesson 047
Evidence named memory result cells

Time 110–160 minutes
Board compile-only Mega replay
Position volatile logical fixture only

Component boundary. `BoundedHomingPolicy` converts copied, already-qualified home and independent-stop evidence into at most one signed semantic step request per accepted frame. It owns no pin, timer, interrupt, bus, sensor, sequencer, coil vector, driver, supply, callback, heap allocation, or physical position.

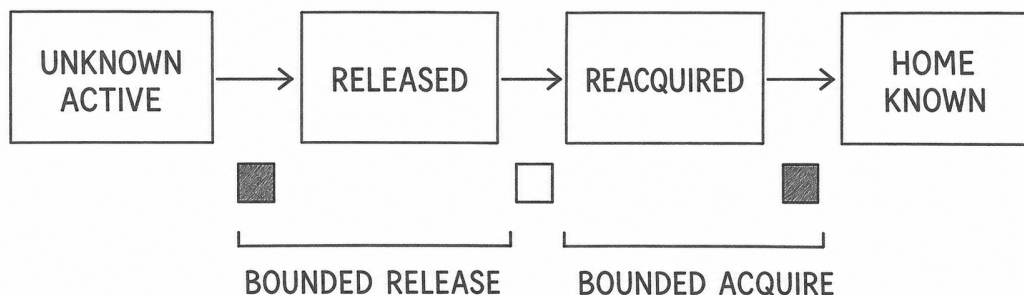
1 Begin with position unknown

Initialization publishes `PositionUnknown`, asserts software stop intent, clears the step request, and establishes no home epoch. A caller must explicitly request homing. Reset, shutdown, restart, source failure, or interrupted logical travel cannot restore an earlier position.

Configuration field	E0 rule
logical bounds	inclusive negative and positive bounds around zero
search direction	exactly -1 or $+1$
release/search limits	separate nonzero step and duration bounds
step cadence	nonzero; delayed calls never burst or catch up
copied evidence	bounded freshness and input skew; stable provenance

The home coordinate is exactly zero. Invalid, unrepresentable, half-range-ambiguous, zero-duration, or internally inconsistent configuration is rejected before the policy starts. These are integer and replay constraints, not mechanical dimensions.

2 Release first, then acquire an edge



Alternative text: a graphite-pencil timeline shows an initially active copied home record moving through bounded release, bounded search, a new qualified inactive-to-active edge, and logical zero. Separate rulers mark release and search limits; no wiring or physical mechanism is depicted.

An active level at startup is ambiguous. It could represent a carriage at home, a broad target encountered elsewhere, reversed polarity, or a stuck sensor. Therefore an initial active record is never accepted as home.

1. Admit one structurally valid, qualified copied home observation.
2. If it is active, enter `SeekingHomeRelease` and request bounded signed steps opposite the configured search direction.
3. Accept only a qualified inactive release from the same evidence domain.
4. Enter `SeekingHome` and request bounded steps in the configured search direction.
5. Accept home only on a new qualified inactive-to-active edge.
6. Clear step intent before publishing position zero and a nonzero home epoch.

An initially inactive observation begins directly in **SeekingHome**. The initial sample, release, and acquisition edge retain the same source kind, source ID, configuration revision, and nonzero qualification epoch. A held active level is not a new edge, and an unqualified inactive value is not a release.

3 Keep release and search bounds independent

Release and approach each have their own issued-step limit and elapsed-time limit. The exact configured boundary is accepted; only the next frame fails. Exhausting release yields **HomeStuckActive**. Exhausting search without the qualified edge yields **HomeNotFound**. Neither retries automatically.

Phase	Accepted evidence	Terminal failure
SeekingHomeRelease	qualified inactive release	stuck active
SeekingHome	qualified inactive-to-active edge	home not found
Homed	bounded target and healthy frames	stop or admitted fault

One accepted update emits at most one due signed request. A delayed call does not create a burst and does not infer skipped physical movement. The policy advances only its own logical fixture coordinate after its request is accepted; it cannot observe an applied motor step.

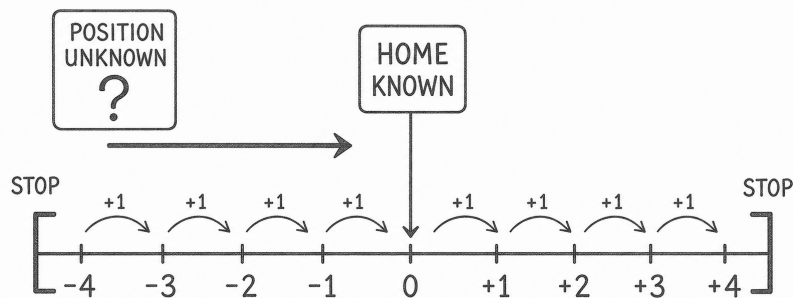
4 Preview the whole transition, then commit once

Every candidate update is side-effect free until committed. `preview()` checks the complete copied frame, command, provenance, time, arithmetic, and transition. `canCommit()` proves that the candidate belongs to the unchanged policy generation. `commit()` then publishes one complete **HomingSnapshot**.

Snapshot field	Meaning at E0
logicalPosition	volatile fixture coordinate
positionKnown	this session accepted a synthetic home edge
stepRequested	one semantic request is due
requestedStepDirection	-1 or +1, never a coil vector
stopIntent	software request to inhibit downstream motion
homingSteps	bounded requests issued in this home attempt
homeEpoch	nonzero identity of the current session-local result

A foreign, stale, changed, replayed, or consumed preview rejects without mutation. Lesson 051 may jointly preflight this semantic candidate with its separately owned Lesson 047 sequencer. Lesson 050 neither contains nor calls that sequencer.

5 Travel only inside the logical excursion



Alternative text: a rough pencil number line places volatile logical zero between inclusive negative and positive bounds. An unknown-state card is separate from the line. The drawn +1 arrows illustrate one-step travel in the positive direction; the policy issues signed -1 or +1 requests and stops at either inclusive boundary.

Only a homed session accepts a target. The target must lie inside the inclusive configured bounds. A zero-distance command completes without a step; otherwise each due accepted frame publishes one signed request until the logical fixture reaches the target. Checked subtraction precedes addition at every signed boundary. The immutable `excursionBounds()` result covers zero and the largest release and search excursions without narrowing before representability is proven. This proves bounded arithmetic, not shaft motion, measured travel, switch activation, or motor shutdown.

6 Let independent stop dominate

A separately admissible, qualified active stop dominates every ordinary request, evidence fault, due step, acquisition edge, final step, and target. It clears the request and asserts stop intent in the same commit, even when unrelated home or command fields are malformed.

If stop interrupts homing or logical travel, the policy enters **Stopped** and forgets position. A stop while idle in **Homed** may retain the known logical coordinate because no displacement was in progress. After release, recovery still requires an explicit command; no held or released level starts motion.

A malformed stop channel cannot be trusted inactive, so complete-frame rejection wins. Other admitted source or motion failures enter **Fault**, assert stop intent, and invalidate the home epoch. Fault recovery requires shutdown followed by initialization. Software stop intent is not an emergency stop, removed motor power, or electrical safe-state proof.

7 Run the deterministic E0 workbook

1. Initialize. Predict unknown position, asserted stop intent, no step, and zero home epoch.
2. Start homing with copied home active. Predict bounded release requests opposite search; repeat an active level and confirm it is not an edge.
3. Supply a qualified inactive release, then due search frames. Predict at most one search request per accepted frame.
4. Supply a new qualified active edge at the exact search bound. Predict **Homed**, logical zero, cleared step request, and a new epoch.
5. Request each inclusive logical bound, then zero distance. Record signed requests and completion without claiming applied motion.
6. Exhaust release and search independently. Distinguish **HomeStuckActive** from **HomeNotFound**.
7. Collide an active valid stop with a due step and malformed unrelated evidence. Predict **Stopped**, no step, and asserted stop intent.
8. Inject provenance change, source fault, same-time changed evidence, rollover ambiguity, restart, and shutdown. Predict atomic rejection or the specified unknown-position terminal state.

E0 result cells

frame	phase	known?	logical position	home epoch
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
source/sequence	home level	requested step	stop intent	status
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

8 Interpret only what E0 observes

Acquisition prediction. Qualified copied evidence with consistent source, revision, epoch, sequence, time, polarity, and status can form the release and acquisition history.

Acquisition observation. Executable host tests inspect the copied records, phase, request direction, logical position, epoch, stop intent, and status in memory.

Acquisition interpretation. A successful replay proves a pure copied-data transition and a session-local fixture origin. It does not prove that a switch, pin, clock, sensor, carriage, or origin was acquired.

Safe-state prediction. Stop, fault, interrupted motion, reset, and shutdown clear the step request, assert stop intent where specified, and invalidate position when displacement cannot be established.

Safe-state observation. Inspect the complete committed snapshot and retained terminal attribution.

Safe-state interpretation. An inert memory result proves only policy precedence. It does not prove de-energized coils, stationary hardware, a dark indicator, or independent removal of actuator power.

9 Keep the resource claim bounded

The canonical Mega replay uses fixed storage, bounded $O(1)$ work, and no heap or hardware resource. Promotion requires the **BoundedHomingPolicy** object to remain at or below 128 bytes (measured: 126 bytes), the sketch to remain within 16,384 bytes flash and 1,024 bytes static SRAM (measured: 8,272 and 574 bytes), at least 448 bytes of conservative stack/ISR allowance, and at least 768 bytes remaining after globals and that allowance.

Compile-only means compile-only. Packaging proves that the public API, fixture, and named result schema fit the Mega build. It does not execute the fixture or prove a powered board populated any result cell.

10 Leave E1 copied-input acceptance blank

E1 remains open. E1 may qualify exact low-energy home and stop producers plus a non-motion observation path. It cannot establish physical home or energize a motor.

Exact sensor and stop specimens

Primary datasheets and revision markings

Supply, logic levels, polarity, and pin order

Qualification, release, edge, and bounce evidence

Named LED/display/test-point observation path

Fault, disconnect, reset, and rollback observation

Reviewer, date, acceptance record

11 Leave E2 powered homing acceptance blank

E2 remains open. No physical build is authorized until the exact sensor, motor, driver, supply, mechanism, protection, and independent power-removal path are identified and reviewed together.

Exact motor, winding, driver, and sensor

Electrically authoritative schematic review

Current-limited actuator supply and common

Measured coil, aggregate, and thermal limits

Restraint, pinch guarding, and overtravel margin

Release, search, stall, and collision measurements

Independent stop and actuator-power removal

Recorded physical homing acceptance result

Reviewer, date, acceptance record

Future E2 work is limited to a lightweight stable tabletop mechanism with no loose load, sharp or high-inertia feature, stored energy, lifting, vehicle, door-lock, launcher, or safety-device role. Firmware stop intent and a software home epoch cannot substitute for restraint or independently removed power.

12 Software evidence and further work

Deterministic tests cover inactive and active starts, bounded release and search, qualification and provenance, exact limits and deadlines, missing and stuck home, bounded targets, preview lifetime, stop and fault collisions, rollover, interruption, recovery, shutdown, and byte-stable replay.

- Canonical compile-only Lesson 050 sketch.
- Deterministic bounded-homing tests.
- Bounded-homing interface.
- Lessons 049–051 design record.
- Deterministic testing contract.
- Safety model.

This untagged print edition has a searchable HTML alternative. It makes no PDF/UA, powered-input, motion, or physical-position claim.